



Semester One Examination, 2025

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section One: Calculator-free

SOLUTIONS

WA student number: In figures

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In words

Your name

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Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	10	10	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free

35% (52 Marks)

This section has **seven** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(7 marks)

- (a) State the size of the matrix $\begin{bmatrix} 22 & 11 & -11 \end{bmatrix}$.

(1 mark)

Solution
Size is 1×3 .
Specific behaviours
✓ correct size

- (b) Evaluate

(i) $4 \begin{bmatrix} 6 & -1 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} -7 & 11 \\ 5 & 2 \end{bmatrix}$.

(2 marks)

Solution
$4 \begin{bmatrix} 6 & -1 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} -7 & 11 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} 24 & -4 \\ 8 & 16 \end{bmatrix} + \begin{bmatrix} -7 & 11 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} 17 & 7 \\ 13 & 18 \end{bmatrix}$
Specific behaviours
✓ correct multiple ✓ correct sum (F/T allowed)

(ii) $\begin{bmatrix} 1 & 7 \\ 4 & 10 \end{bmatrix} \times \begin{bmatrix} 3 & -2 \\ 2 & 1 \end{bmatrix}$.

(2 marks)

Solution
$\begin{bmatrix} 1 & 7 \\ 4 & 10 \end{bmatrix} \times \begin{bmatrix} 3 & -2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 5 \\ 32 & 2 \end{bmatrix}$
Specific behaviours
✓ one correct element of product ✓ correct product

- (c) Determine the value of p and the value of q given $\begin{bmatrix} p \\ -4 \end{bmatrix} - 3 \begin{bmatrix} 7 \\ q \end{bmatrix} = \begin{bmatrix} -5 \\ 2 \end{bmatrix}$.

(2 marks)

Solution
$p - 21 = -5 \Rightarrow p = 16$
$-4 - 3q = 2 \Rightarrow q = -2$
Specific behaviours
✓ correct value of p ✓ correct value of q

Question 2

(7 marks)

(a) Evaluate the following expressions:

(i) $0.25 \times 20 - 8$.

(1 mark)

Solution
$0.25 \times 20 - 8 = 5 - 8 = -3$
Specific behaviours
✓ correct value

(ii) $\frac{3}{0.5 \times 8} - \frac{1}{6^2 - 32}$.

(2 marks)

Solution
$\frac{3}{0.5 \times 8} - \frac{1}{6^2 - 32} = \frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2} = 0.5$
Specific behaviours
✓ simplifies either fraction ✓ correct value

(b) Determine the value of $\sqrt{5b - 3ac - 2}$ when $a = -3$, $b = 6$ and $c = 4$.

(2 marks)

Solution
$\begin{aligned}\sqrt{5b - 3ac - 2} &= \sqrt{5 \times 6 - 3 \times (-3) \times 4 - 2} \\ &= \sqrt{30 + 36 - 2} \\ &= \sqrt{64} = 8\end{aligned}$
Specific behaviours
✓ shows any correct simplification ✓ correct value

(c) Given that $x^2 - y^2 = (x + y)(x - y)$, evaluate $41^2 - 11^2$.

(2 marks)

Solution
$\begin{aligned}41^2 - 11^2 &= (41 + 11)(41 - 11) \\ &= 52 \times 30 \\ &= 1560\end{aligned}$
Specific behaviours
✓ substitutes and simplifies each bracket ✓ correct value

Question 3

(8 marks)

In matrix $X = \begin{bmatrix} 5 & 7 & 8 \\ 6 & 7 & 6 \end{bmatrix}$, element $x_{i,j}$ is the number of single people ($i = 1$) or couples ($i = 2$) who attended a dance class on night j .

- (a) Show that a total of 17 people attended the dance class on night 1. (1 mark)

Solution
5 single people plus 6 couples is $5 + 2 \times 6 = 5 + 12 = 17$ people in total.
Specific behaviours
✓ shows how to obtain total number of people

- (b) Determine $[1 \ 2] \times X$ and explain what information is contained in your answer. (2 marks)

Solution
$[1 \ 2] \times \begin{bmatrix} 5 & 7 & 8 \\ 6 & 7 & 6 \end{bmatrix} = [17 \ 21 \ 20]$
The matrix shows the number of people who attended the dance class on each night.
Specific behaviours
✓ correct matrix product
✓ correct explanation (must mention each night)

The fee to attend a dance class for one night was \$20 for a single and \$30 for a couple.

- (c) Represent the fees in row matrix Y so that the matrix product $Y \times X$ will have a meaningful result. (1 mark)

Solution
$Y = [20 \ 30]$
Specific behaviours
✓ correct matrix

- (d) Determine $Y \times X \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. (3 marks)

Solution
$Y \times X \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = [20 \ 30] \times \begin{bmatrix} 5 & 7 & 8 \\ 6 & 7 & 6 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = [20 \ 30] \times \begin{bmatrix} 20 \\ 21 \\ 19 \end{bmatrix} = [970]$
or
$Y \times X \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = [20 \ 30] \times \begin{bmatrix} 5 & 7 & 8 \\ 6 & 7 & 6 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = [280 \ 350 \ 340] \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = [970]$
Specific behaviours
✓ at least one correct element seen in attempt to multiply a pair of matrices
✓ one correct matrix product seen
✓ correct result

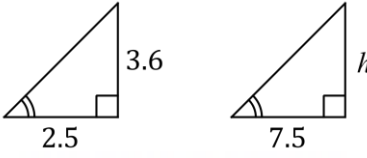
- (e) Explain what information is contained in the matrix calculated in part (d). (1 mark)

Solution
Total attendance fees for the three nights.
Specific behaviours
✓ correct explanation

Question 4

(7 marks)

- (a) A vertical flagpole, of height 3.6 metres, stands near a vertical lighting tower on a level sports ground. At the instant the length of the shadow of the flagpole is 2.5 metres, the length of shadow of the lighting tower is 7.5 metres. By first sketching the two similar triangles involved in this situation, determine the height of the lighting tower. (3 marks)

Solution	
 Scale factor of similar triangles is $7.5 \div 2.5 = 3$. Hence height of tower is $h = 3.6 \times 3 = 10.8$ metres.	
Specific behaviours	
✓ sketch of similar triangles (might overlap) ✓ indicates scale factor ✓ correct height of tower	

- (b) A measurement of 60 millimetres on a construction plan represents a length of 12 metres in a building. Determine the scale factor of the plan in the form 1: n . (2 marks)

Solution
60 mm: 12 m = 60: 12 000 = 6: 1200 = 1: 200
Specific behaviours
✓ writes a correct ratio after eliminating units ✓ correct scale factor

- (c) The straight-line distance between points P and Q on a map drawn to a scale of 1: 25 000 is 10 centimetres. Determine the actual straight-line distance between the points in kilometres. (2 marks)

Solution
Actual distance is $10 \times 25\,000 = 250\,000$ cm.
250 000 cm is 2500 metres which is 2.5 kilometres.
Specific behaviours
✓ indicates correct distance in centimetres ✓ correct distance in kilometres

Question 5

(7 marks)

Takeshi lives in a boarding house that charges him \$420 each week for food and lodging.

He has allocated all his weekly income of \$840 to one of five budget categories shown at right.

He always allocates 15% of his weekly income to savings.

	A	B
1	Board	\$420
2	Technology	\$55
3	Car	
4	Savings	
5	Other	\$99
6	Total	\$840

- (a) The boarding house fines residents 5% of their weekly charge for late payment. Determine how much Takeshi would be fined for late payment.

(1 mark)

Solution
$\$420 \times 10\% = \42 , $\$420 \times 5\% = \21 Fine is \$21.
Specific behaviours
✓ correct amount

- (b) Determine his budget allocation for savings.

(1 mark)

Solution
$\$840 \times 10\% = \84 , $\$840 \times 5\% = \42 Allocation for savings is $\$84 + \$42 = \$126$.
Specific behaviours
✓ correct amount

- (d) Takeshi plans to increase the budget allocation for Technology by \$12 and for **Other** by the same amount. When he does so, which of these two categories will have the largest percentage increase? Justify your answer.

(2 marks)

Solution
Technology.
$\frac{12}{55}$ is a larger fraction than $\frac{12}{99}$, and so $\frac{12}{55}$ will also be a larger percentage.
Specific behaviours
✓ states correct category ✓ justifies answer

Takeshi estimates that 40% of his car budget is spent on fuel.

- (c) Assuming a year of exactly 52 weeks, determine his annual fuel costs.

(3 marks)

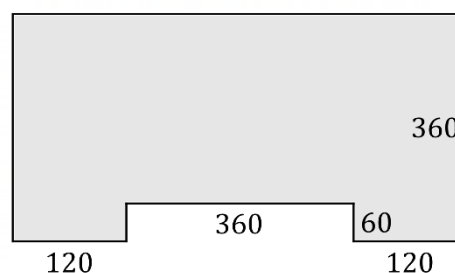
Solution
$\$420 + \$55 + \$126 + \$99 = \$700$ Car budget is $\$840 - \$700 = \$140$. 40% of \$140 is \$56 Annual cost is $52 \times \$56 = \2912
Specific behaviours
✓ calculates amount budgeted for car ✓ calculates 40% of budgeted amount ✓ correct annual fuel cost (F/T allowed)

Question 6**(8 marks)**

The floor plan of a room, not drawn to scale, is shown. All corners are right-angled and all measurements are in centimetres.

The floor of the room is to be tiled with rectangular tiles that measure 0.3 metres by 0.6 metres.

The tiles are sold in boxes of 10 at a cost of \$90 per box. Only full boxes can be bought.



- (a) How long is the longest side of the room, in metres?

(1 mark)

Solution
Length is $1.2 + 3.6 + 1.2 = 6$ metres.
Specific behaviours
✓ correct length

- (b) Determine how much it will cost to buy tiles for the room.

(5 marks)

Solution
LH part of room (360×120) needs $6 \times 4 = 24$ tiles.
Central part of room (300×360) needs $5 \times 12 = 60$ tiles.
Total number of tiles will be $24 + 60 + 24 = 108$.
Number of boxes needed is $108 \div 10 = 10 \text{ r}8$, so 11 full boxes.
Cost of tiles is $11 \times \$90 = \990 .
Specific behaviours
✓ indicates partitioning of floor area
✓ correct number of tiles for one partitioned area
✓ correct number of tiles for floor
✓ indicates number of boxes required
✓ correct cost of tiles
Correct alternate solution is allowed

Once the floor has been tiled, a thin wooden strip that costs \$5 per metre is to be glued around the edge of the tiled area.

- (c) Determine the cost of wooden strip to edge the tiles.

(2 marks)

Solution
Perimeter is $6 + 6 + 3.6 + 3.6 + 0.6 + 0.6 = 20.4$ metres.
Cost is $20.4 \times \$5 = \102 .
Specific behaviours
✓ correct length of wooden strip required
✓ correct cost

Question 7

(8 marks)

Fruit pickers at a farm are paid \$10 per carton of pears plus a wage of \$20 per hour, with both rates increasing by 50% on weekends.

- (a) Determine how much Amana will earn for picking 12 cartons of pears during an 8 hour shift on a Wednesday. (2 marks)

Solution
Cartons: $\$10 \times 12 = \120
Wage: $\$20 \times 8 = \160
Total: $\$120 + \$160 = \$280$
Specific behaviours
✓ correct earnings from cartons or wages
✓ correct earnings

- (b) Determine how much Babbe will earn for picking 8 cartons of pears during a 6 hour shift on a Saturday. (2 marks)

Solution
Cartons: $\$10 \times 1.5 \times 8 = \120
Wage: $\$20 \times 1.5 \times 6 = \180
Total: $\$120 + \$180 = \$300$
Specific behaviours
✓ correct earnings from cartons or wages
✓ correct earnings

- (c) Carma picked 8 cartons of pears during a 5 hour shift on Sunday and 11 cartons during a second shift on Monday. Given that she earned \$520 for the two days' work, determine how long she worked for on Monday. (4 marks)

Solution
Sunday earnings: $\$10 \times 8 \times 1.5 + \$20 \times 5 \times 1.5 = \$120 + \$150 = \270
Monday cartons: $\$10 \times 11 = \110
Monday wages: $\$520 - \$270 - \$110 = \140
Monday hours: $\$140 \div \$20 = 7$ hours
Specific behaviours
✓ calculates Sunday earnings
✓ calculates earnings from cartons on Monday
✓ calculates wages for Monday
✓ correct Monday hours worked (F/T allowed)

Supplementary page

Question number: _____

Supplementary page

Question number: _____

